Summary in Graph

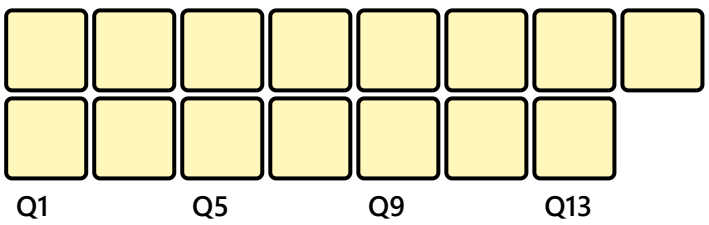
# Exam Summary (GO Classes CS/DA Test Series 2026 | Probability and Statistics | Bayes Theorem, Conditional Probability | Topic Wise Test)

| | | | |
|---|---|---|---|
| Qs. Attempted: | 0 (0 + 0) | Correct Marks: | 0 (0 + 0) |
| Correct Attempts: | 0 (0 + 0) | Penalty Marks: | 0 (0 + 0) |
| Incorrect Attempts: | 0 (0 + 0) | Resultant Marks: | 0 (0 + 0) |

| | |
|---|---|
| Total Questions: | 15 (5 + 10) |
| Total Marks: | 25 (5 + 20) |
| Exam Duration: | 45 Minutes |
| Time Taken: | 0 Minutes |

## Individual Question Statistics

Select Metric: Marks Obtained




* Average Stats based on first-attempt stats of Top 10% (159) out of 1588 users

EXAM RESPONSE EXAM STATS FEEDBACK

## Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Probability

A gambler has in his pocket a fair coin and a two-headed coin. He selects one of the coins at random and flips it. If the coin comes up heads, what is the probability that it is the fair coin?

A. $\frac{1}{2}$

B. $\frac{1}{3}$

C. $\frac{2}{3}$

D. $\frac{3}{4}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 04sec | Discuss

**Q #2** Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

In a probabilistic model, it is given that $\mathrm{P}[A] = \mathrm{a}, \mathrm{P}[B] = \mathrm{b}$, and $\mathrm{P}\left[(B^c \mid A^c)\right] = \mathrm{e}$. Determine $\mathrm{P}[B \mid A]$ in terms of $\mathrm{a}, \mathrm{b}$, and $\mathrm{e}$.

A. $\mathrm{P}[B \mid A] = \frac{\mathrm{b}-(1-\mathrm{e})(1-\mathrm{a})}{\mathrm{a}}$
B. $\mathrm{P}[B \mid A] = \frac{\mathrm{b}+(1-\mathrm{e})(1-\mathrm{a})}{\mathrm{a}}$
C. $\mathrm{P}[B \mid A] = \frac{\mathrm{b}+(1-\mathrm{e})(1+\mathrm{a})}{\mathrm{a}}$
D. $\mathrm{P}[B \mid A] = \frac{\mathrm{b}-(1+\mathrm{e})(1-\mathrm{a})}{\mathrm{a}}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #3** Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Mark all expressions which indicate that $\mathrm{X}$ is independent of $\mathrm{Y}$ given $\mathrm{Z}$.

A. $P(X, Y \mid Z) = P(X \mid Z)P(Y \mid Z)$
B. $P(X, Y, Z) = P(X, Z)P(Y)$
C. $P(X \mid Y, Z) = P(X \mid Z)$
D. None of the above

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #4** Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Select all expressions that are equivalent to $P(A, B \mid C)$,
given that $A \perp B \mid C$,
where $\perp$ denotes conditional independence — that is,
*A is independent of B when C is known.*

A. $\frac{P(A|C)P(C|B)P(B)}{P(B,C)}$

B. $\frac{P(A|C)P(B,C)}{P(C)}$

C. $P(A \mid B, C)P(B \mid C)$

D. $\frac{P(C|A,B)P(B|A)P(A)}{P(B|C)P(C)}$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #5** Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Select all expressions that are equivalent to $\mathbf{P}(\mathbf{A} \mid \mathbf{B}, \mathbf{C})$, given that A ⊥ B | C

A. $\frac{P(C|A)P(A|B)P(B)}{P(C)}$

B. $\frac{P(B,C|A)P(A)}{P(B,C)}$

C. $\frac{P(A|C)P(C|B)P(B)}{P(B,C)}$

D. $\frac{P(C|A,B)P(B|A)P(A)}{P(B|C)P(C)}$

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #6** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose that the probability of making a mistake on your homework depends on how much sleep you got the previous night. The probabilities are as follows:

- If you got at least $8$ hours of sleep, there is a $10\%$ chance of making a mistake.
- If you got between $6$ and $8$ hours of sleep, there is a $40\%$ chance of making a mistake.
- If you got less than $6$ hours of sleep, there is a $75\%$ chance of making a mistake.

Additionally, the probabilities for the amount of sleep are:

- There is a $20\%$ chance that you got at least $8$ hours of sleep.
- There is a $50\%$ chance that you got between $6$ and $8$ hours of sleep.

What is the overall probability that you make a mistake on your homework?

A. $0.265$

B. $0.365$

C. $0.445$

D. $0.545$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #7** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A gambler has in his pocket a fair coin and a two-headed coin. He selects one of the coins at random, and then he flips it. What is the probability that the flip results in heads?

A. $0.5$

B. $0.75$

C. $0.25$

D. $1.0$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

In a game where the probability of winning on a single toss of a dice is $p$, Player $A$ starts and if he fails, he passes the dice to Player $B$. They continue tossing the dice alternately until one of them wins. What is the probability that Player $A$ wins?

A. $\frac{1}{p}$

B. $\frac{1}{1-(1-p)^2}$

C. $\frac{1}{2-p}$

D. $\frac{p}{2}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Consider the following statements, where $P(\cdot)$ denotes probability and assuming all conditional probabilities involved are non-zero:
S1: If $P(a \mid b, c) = P(b \mid a, c)$, then $P(a \mid c) = P(b \mid c)$.
S2: If $P(a \mid b) = P(a)$, then $P(b \mid c) = P(b)$.
Choose the correct option that correctly interprets the implications of these statements:

A. Statement $\mathrm{S1}$ is true, and statement $\mathrm{S2}$ is false.
B. Statement $\mathrm{S1}$ is false, and statement $\mathrm{S2}$ is true.
C. Both statement $\mathrm{S1}$ and statement $\mathrm{S2}$ are true.
D. Both statement $\mathrm{S1}$ and statement $\mathrm{S2}$ are false.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Which of the following statements is/are TRUE?

A. For any three events $A$, $B$, and $C$ where $A$ and $B$ are disjoint and $0 < P(C) < 1$, $P(A \cup B \mid C) = P(A \mid C) + P(B \mid C)$
B. For any two events $A$ and $B$ where $0 < P(B) < 1$, $P(A \mid B) = 1 - P(A \mid B^c)$
C. For any three events $A$, $B$, and $C$, $P(A \cap B \cap C) = P(A)P(B)P(C)$
D. For any two events $A$ and $B$, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Given events $A_1, A_2, A_3$ in a sample space $S$ with probabilities $P(A_1) = 0.4, P(A_2) = 0.2, P(A_3) = 0.4$, and assuming $A_i \cap A_j = \emptyset$ for $i \neq j$, and another event $B$ with conditional probabilities $P(B \mid A_1) = 0.1, P(B \mid A_2) = 0.2, P(B \mid A_3) = 0.05$, what is the probability that either $A_1$ or $A_2$ occurs

given that $B$ has occurred?

A. $P(A_1 \cup A_2 \mid B) = 0.2$
B. $P(A_1 \cup A_2 \mid B) = 0.5$
C. $P(A_1 \cup A_2 \mid B) = 0.8$
D. $P(A_1 \cup A_2 \mid B) = 0.9$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Consider a scenario where a ball is randomly placed in one of $n$ boxes, each with probability $P_i$ for box $i$. If the ball is in box $i$, a search of that box will uncover it with probability $\alpha_i$. The conditional probability that the ball is in box $j$, given that a search of box $i$ did not uncover it, is:

A. $P($ ball in box $j|$ not found in box $i) = \frac{P_j}{1-\alpha_i P_i}$, if $j \neq i$

B. $P($ ball in box $j|$ not found in box $i) = \frac{(1-\alpha_i)P_i}{1-\alpha_i P_i}$, if $j \neq i$

C. $P($ ball in box $j|$ not found in box $i) = \frac{1-\alpha_i P_i}{P_j}$, if $j \neq i$

D. $P($ ball in box $j|$ not found in box $i) = \frac{P_i}{1-\alpha_i}$, if $j \neq i$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

You have three coins in your pocket: two fair ones and one biased with probability $p$ for heads and $1-p$ for tails.
Let $F, U$, and $H$ denote the events that:

- $F$ : The selected coin is fair.
- $U$ : The selected coin is unfair.
- $H$ : The coin lands heads up.

What is the probability that the selected coin is one of the fair coins, given that it lands heads up? (i.e., $P(F \mid H)$ )

A. $\frac{1}{1+p}$

B. $\frac{p}{1+p}$

C. $\frac{1}{2}$

D. $\frac{2}{3}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Is it possible to have events $A, B, C$ such that:

$$P(A \mid C) < P(B \mid C) \text{ and } P(A \mid C^c) < P(B \mid C^c), \text{ yet } P(A) > P(B) \text{ ?}$$

A. Yes, it is possible, but only if $P(C) = P(C^c)$.
B. Yes, it is always possible, irrespective of $P(C)$ being equal to $P(C^c)$.
C. No, it is not possible; such a scenario contradicts probability rules.
D. The answer cannot be determined from the information given.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Given probabilities:

$$\mathbb{P}(A) = \frac{1}{5}, \quad \mathbb{P}(B) = \frac{3}{10}, \quad \mathbb{P}(A \cup B) = \frac{11}{25}$$

and assuming events $A$ and $B$ are independent, what is $\mathbb{P}(A \cap B \mid A \cup B)$ ?

A. $\frac{3}{22}$

B. $\frac{11}{25}$

C. $\frac{1}{5}$

D. $\frac{2}{5}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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